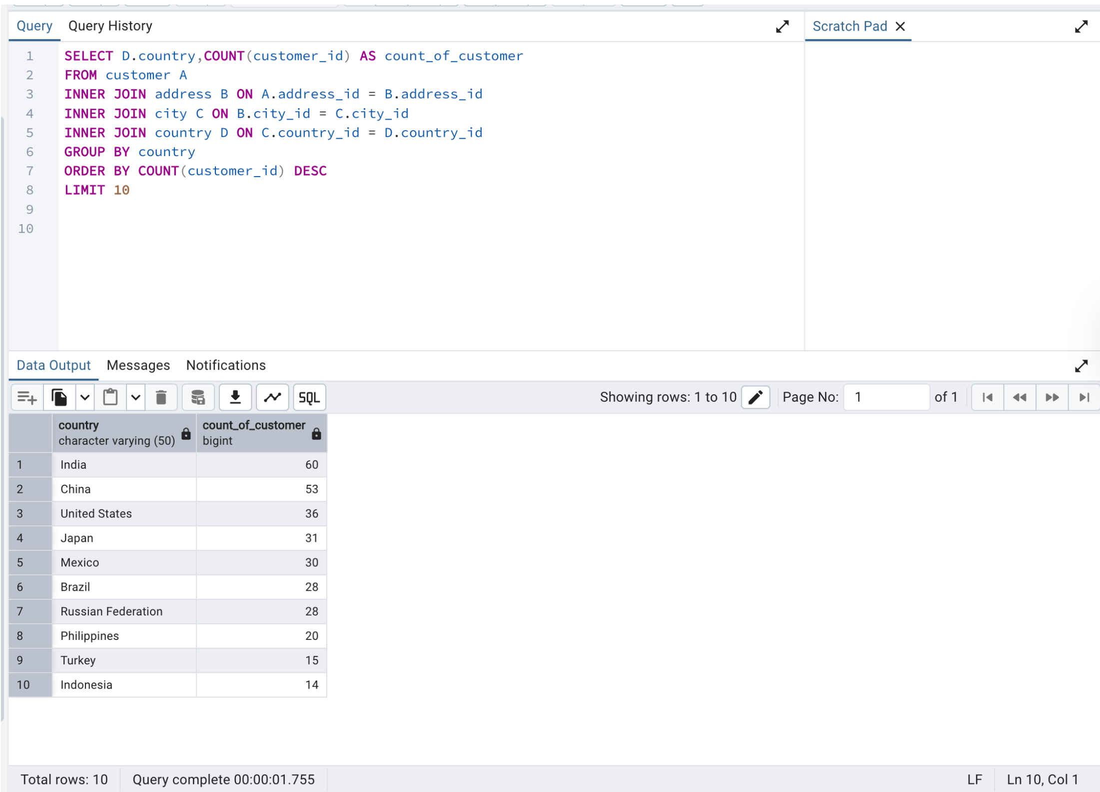
Exploratory Data Analysis

Finding the top 10 countries for Rockbuster in terms of customer numbers.

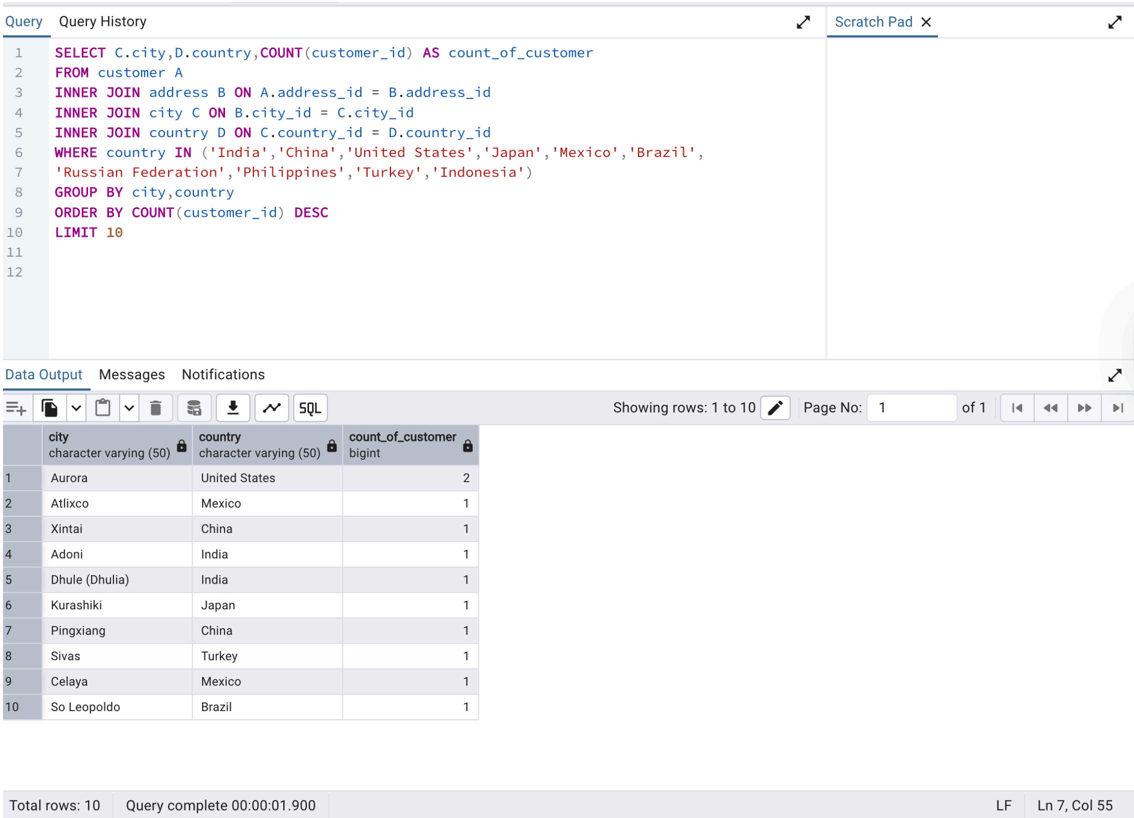
SELECT D.country,COUNT(customer_id) AS count_of_customer
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
GROUP BY country
ORDER BY COUNT(customer_id) DESC
LIMIT 10



I first decided to find the concerned tables for this query and noted down their relationship with each other. The concerned tables are customer, address, city and country. Although I want the information from only two tables (customer and country), I cannot because they are not directly linked with a key. Hence, I used an inner join to establish a relationship between the tables via primary/foreign keys. The tables are connected as follows:
Customer (address_id) → Address (address_id)
Address (city_id) → City (city_id)
City (country_id) → Country (country_id)
I have grouped the results by country since we wanted the top 10 countries where Rockbuster has the maximum customers.

Identifying the top 10 cities that fall within the top 10 countries I identified in step 1.

SELECT C.city,D.country,COUNT(customer_id) AS count_of_customer
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
WHERE country IN ('India','China','United States','Japan','Mexico','Brazil',
'Russian Federation','Philippines','Turkey','Indonesia')
GROUP BY city,country
ORDER BY COUNT(customer_id) DESC
LIMIT 10




```sql
SELECT C.city,D.country,COUNT(customer_id) AS count_of_customer
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
WHERE country IN ('India','China','United States','Japan','Mexico','Brazil',
'Russian Federation','Philippines','Turkey','Indonesia')
GROUP BY city,country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
```

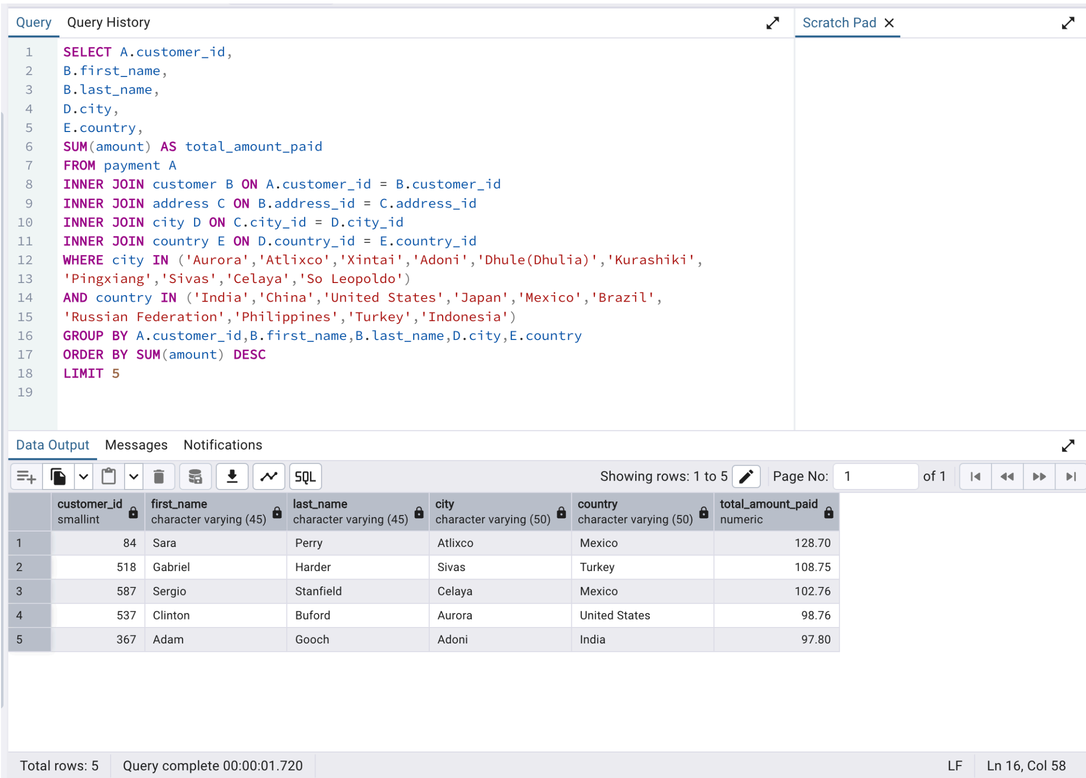
	city character varying (50)	country character varying (50)	count_of_customer bigint
1	Aurora	United States	2
2	Atlixco	Mexico	1
3	Xintai	China	1
4	Adoni	India	1
5	Dhule (Dhulia)	India	1
6	Kurashiki	Japan	1
7	Pingxiang	China	1
8	Sivas	Turkey	1
9	Celaya	Mexico	1
10	So Leopoldo	Brazil	1

Since I now know the top 10 countries, I put them in the WHERE clause. I also selected the city name along with the country name to display in my output. The other change I made in this query was to group it by both city and country.

Finding the top 5 customers from the top 10 cities who've paid the highest total amounts to Rockbuster. The customer team would like to reward them for their loyalty!

SELECT A.customer_id,
B.first_name,
B.last_name,
D.city,
E.country,
SUM(amount) AS total_amount_paid

FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
'Pingxiang','Sivas','Celaya','So Leopoldo')
AND country IN ('India','China','United States','Japan','Mexico','Brazil',
'Russian Federation','Philippines','Turkey','Indonesia')
GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
ORDER BY SUM(amount) DESC
LIMIT 5



For this query, we have 5 tables (payment, customer, address, city and country). We'll be using two WHERE clauses specifying both the city and country.